

DataLyzer

Statistical Analysis Report

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Source: sample_data.csv

1. Dataset Overview

Original rows	315
Processed rows	300
Rows removed	15
Columns	6
Numeric columns	5
Categorical cols	1
Missing values (post-clean)	0

Numeric columns: age, income, score, spend, label

Categorical columns: region

2. Preprocessing Steps

- Removed 15 duplicate row(s).
- Filled missing values using 'mean' strategy.
- Z-score normalization applied (target mean=0.0, std=1.0).

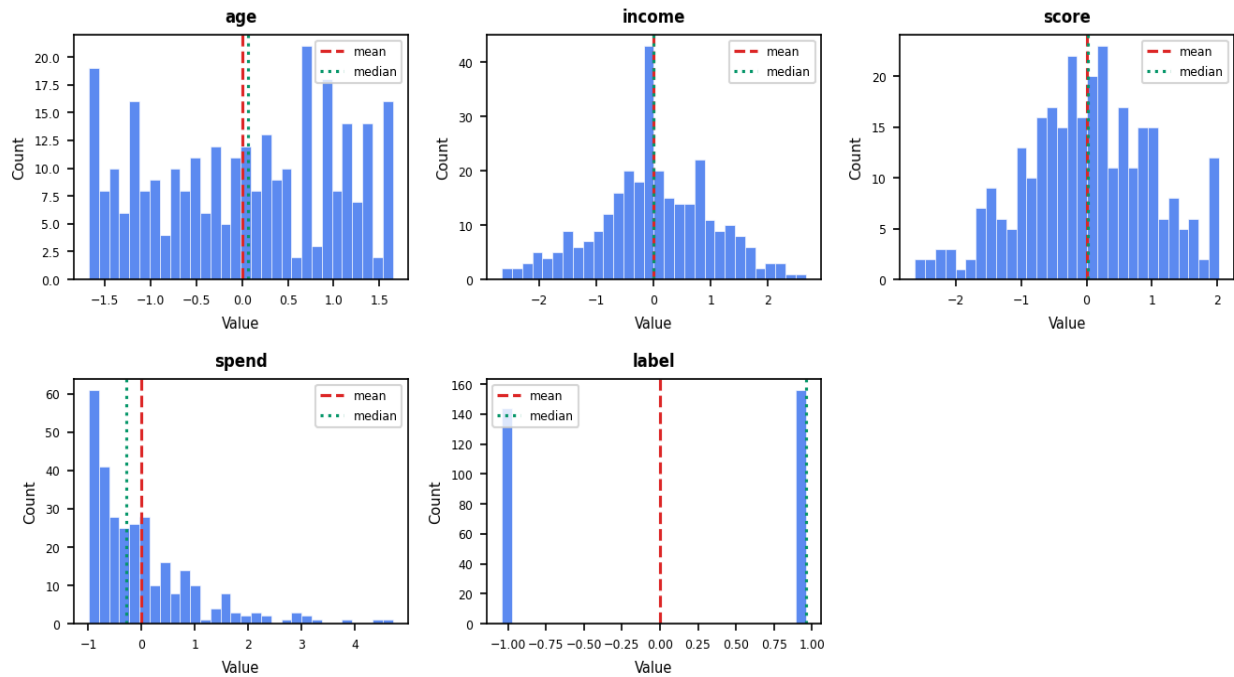
3. Data Preview (first 10 rows)

age	income	score	spend	label	region
1.1927	0.2917	0.0911	-0.0039	0.9592	West
0.4375	0.8206	1.0139	0.0914	0.9592	North
-0.6197	-1.0327	-0.8915	0.7491	0.9592	South
1.4947	0.0061	0.141	-0.7306	0.9592	West
-1.1484	-1.5475	1.5589	-0.3755	0.9592	North
-0.1666	-0.9345	-2.3489	-0.7649	0.9592	North
1.1927	0.7572	-0.7327	3.0581	0.9592	East
-0.3177	-0.389	-0.825	1.6637	-1.0391	East

age	income	score	spend	label	region
-0.0156	1.0962	-0.1205	-0.9584	-1.0391	West
-0.9218	0.408	0.016	0.1322	-1.0391	South

4. Feature Distributions

Feature Distributions



5. Measures of Central Tendency

Column	Mean	Median	Mode	N
age	0.0000	0.0599	0.0599	300
income	-0.0000	0.0000	0.0000	300
score	-0.0000	0.0163	2.0264	300
spend	0.0000	-0.2799	-0.9895	300
label	0.5200	1.0000	1.0000	300

6. Measures of Dispersion

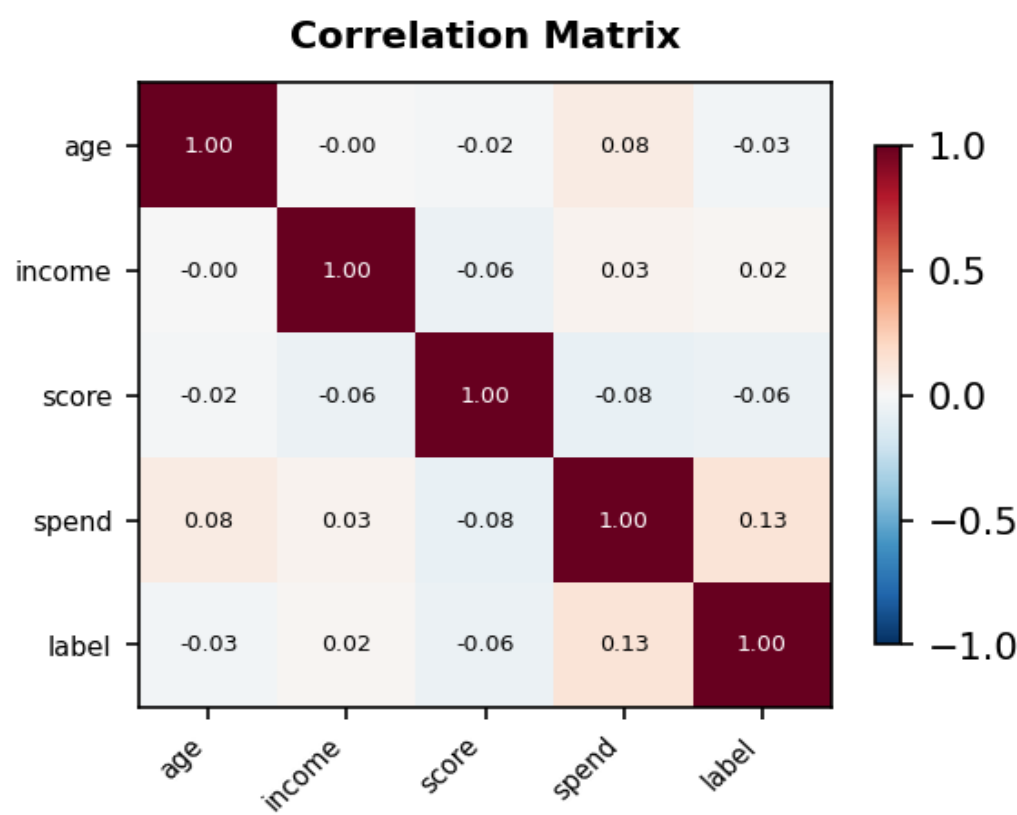
Column	Std Dev	Variance	Range	IQR	Min	Max
age	1.0000	1.0000	3.3228	1.8124	-1.6770	1.6458

income	1.0000	1.0000	5.3162	1.3288	-2.6569	2.6593
score	1.0000	1.0000	4.6542	1.3329	-2.6279	2.0264
spend	1.0000	1.0000	5.6953	1.1507	-0.9895	4.7058
label	0.5004	0.2504	1.0000	1.0000	0.0000	1.0000

7. Distribution Shape (Skewness & Kurtosis)

Column	Skewness	Kurtosis	Skew Interpretation
age	-0.0914	-1.2137	Symmetric (left)
income	-0.1346	-0.0686	Symmetric (left)
score	-0.1212	-0.2504	Symmetric (left)
spend	1.7537	3.7260	High skew (right)
label	-0.0805	-2.0069	Symmetric (left)

8. Correlation Matrix



9. ROC-AUC Analysis

Feature	AUC Score	Interpretation
age	0.4847	Poor (<0.6)
income	0.5067	Poor (<0.6)
score	0.4733	Poor (<0.6)
spend	0.5717	Poor (<0.6)

